BusSetu

"Real-Time Bus Tracking and Management System"

SRS on

Mini Project-IV

(UCBIL0671)

by

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KMT Live Bus Tracking System

Project Description

The **Live Bus Tracking System** is a digital platform designed to provide real-time tracking of public transport buses. The system allows citizens to track the live location of buses, view routes, estimate arrival time (ETA), and select their source and destination stops. The system improves public transport efficiency by providing transparency and reducing passenger waiting time.
The platform includes three main users: **Admin, Driver, and Citizen**. The admin manages routes, stops, buses, and drivers. Drivers use a mobile application to start trips and send GPS data. Citizens can track buses in real-time without logging in.

Functional Requirements

Module 1: User Authentication
Function Name: User Registration and Login
Description:
Allows admin and driver users to register and securely log into the system using email and password. Authentication is handled using encrypted passwords and JWT tokens.
Input:
- Name
- Email
- Password
- Role (Admin / Driver)

Expected Output:
- User account created successfully
- JWT authentication token generated on login

Module 2: Route Management

Function Name: Create and Manage Routes
Description:
Admin can create bus routes manually or upload route details using Excel files. Each route consists of multiple stops arranged in sequence.
Input:
- Route name
- Excel file containing route data (optional)

Expected Output:
- New route added to database
- Routes available for buses and citizens

Module 3: Stop Management
Function Name: Add Stops to Routes

Description:
Admin can define stops between the starting and ending points of a route. Each stop contains geographic coordinates and sequence order.
Input:
- Stop name
- Latitude
- Longitude
- Route ID
- Stop order

Expected Output:
- Stops successfully linked to the route
- Stops displayed to citizens for route selection

Module 4: Bus Management

Function Name: Bus Registration
Description:
Admin can add buses to the system and make them available for drivers to select during trip initialization.
Input:
- Bus number
- Bus status

Expected Output:
- Bus added to the system
- Bus available for driver selection

Module 5: Trip Initialisation

Function Name: Start Bus Trip
Description:
Driver logs in and selects a bus and route before starting the journey. The system creates a trip record and begins live tracking.
Input:
- Driver login credentials
- Bus ID
- Route ID

Expected Output:
- Trip created successfully
- Live tracking session started

Module 6: Live Bus Tracking
Function Name: GPS Location Update

Description:
Driver's mobile application continuously sends GPS location data to the backend server during the trip.
Input:
- Trip ID
- Latitude
- Longitude
- Speed

Expected Output:
- Location stored in database
- Real-time bus location available for users

Module 7: Citizen Bus Tracking
Function Name: Bus Search and ETA
Description:
Citizens select their source and destination stops to view available buses and track their movement on the map.
Input:
- Source stop
- Destination stop

Expected Output:
- List of active buses
- Bus location displayed on map
- Estimated time of arrival (ETA)

Non-Functional Requirements
Security
The system uses **JWT authentication** and encrypted passwords to ensure secure access for admin and driver accounts.
Performance
The system processes location updates efficiently to provide near real-time tracking without significant delays.
Scalability
The architecture supports expansion to multiple routes, buses, and cities without major redesign.
Reliability
The system ensures stable operation even with frequent GPS updates from drivers.
Usability
The interface is designed to be simple and intuitive so that citizens can easily track buses without needing an account.
Portability
The system can run on different platforms including web browsers and Android mobile devices.
Maintainability

The backend uses modular architecture, making it easier to update or expand system features.

Hardware Requirements

Server Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 20 GB free space

Client Requirements

- Smartphone with GPS capability (for drivers)
- Internet connectivity

Citizen Access

- Smartphone or computer
- Internet browser

Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Backend:** Node.js with Express.js
- **Frontend:** React.js
- **Mobile App:** Kotlin (Android)
- **Database:** PostgreSQL
- **APIs:** REST APIs
- **Map Services:** Google Maps API / OpenStreetMap